

Deep Convolutional Neural Networks for Fine-Grained Spice Image Classification

Abstract: Spices are an essential part of cooking traditions around the world, but identifying them correctly can be difficult because many spices look very similar, share overlapping colors and textures, and appear in different forms such as whole, ground, or dried. In this project, we develop a deep learning based spice classification system using Convolutional Neural Networks, where texture and color features are combined to better capture the visual details of each spice. The system is designed to classify more than 30 common spice types with high accuracy. To reduce confusion between similar looking spices such as cumin and caraway or paprika and chili powder, we introduce a contrastive fine tuning step along with a spice specific data augmentation approach, which helps the model learn more distinctive features and improve performance. The proposed system is intended for practical use in applications such as smart kitchen assistants, food quality control, and retail inventory management.

Introduction:

Problem Statement : Accurate spice identification remains a challenging task in computer vision due to high visual similarity among spice categories. Many commonly used spices share overlapping color distributions (e.g., brown, red, yellow) and similar texture patterns, particularly in powdered form, making fine-grained classification difficult even for human observers. Existing image classification models often underperform in this domain due to limited availability of spice-specific datasets and their inability to capture subtle micro-textural differences essential for distinguishing visually similar spices.

Motivation: The growing adoption of smart kitchen systems, increasing demand for food traceability, and advancements in AI-assisted food analysis highlight the need for automated spice recognition. Current food recognition datasets either exclude spices or represent them at a coarse level, limiting their practical

applicability. This creates a research gap for developing a specialized, fine-grained spice classification framework with both scientific and commercial relevance.

Objective: The primary objective of this work is to develop a robust deep learning-based spice classification system capable of accurately recognizing over 30 spice categories. The model aims to outperform general-purpose architectures on spice-specific benchmarks, ensure robustness across different presentation forms (whole, ground, and dried), and provide interpretability through visualization techniques such as Grad-CAM.

Why Our Approach is Better: Unlike conventional food classification approaches that treat spices as a secondary category, this work focuses specifically on spice recognition. We construct a dedicated high-resolution dataset under diverse conditions, apply spice-specific data augmentation techniques, and incorporate a contrastive fine-tuning strategy to better distinguish visually similar classes. This targeted approach improves feature discrimination and is expected to achieve significant performance gains over standard fine-tuning methods.

Dataset:

Primary Dataset: Spice Image Dataset (custom-curated): ~11,000 images across 11 spice classes collected from open-source repositories (Open Images, iNaturalist) and self-captured images. Each class contains 1000 images in varied lighting, backgrounds, and presentation forms (whole, ground, raw).

Dataset Link:

<https://www.sciencedirect.com/science/article/pii/S2352340925008194>

Supplementary Datasets:

- SPICE-30 Benchmark (Kaggle): baseline comparison dataset
- Food-101 (spice subset): transfer learning pre-training
- Custom macro-photography subset (~2,000 images) for texture analysis

Data Split: Train: 70% | Validation: 10% | Test: 20%

Stratified split ensuring all presentation forms represented in each split.

Preprocessing: Resize to 512 x 512, normalize with ImageNet mean/std, background removal experiments using SAM (Segment Anything Model) to isolate spice regions, augmentation: horizontal/vertical flip, rotation ($\pm 30^\circ$), color jitter (calibrated to spice palette), Gaussian blur.

Methodology:

Model Architecture:

- Proposed: Multi-Modal Fusion CNN
- Backbone: EfficientNet-B4 (pre-trained on ImageNet) for global feature extraction
- Texture Branch: Local Binary Pattern (LBP) + GLCM features fed into a shallow MLP(Multi Layer Perception)
- Color Branch: HSV histogram features (spice-calibrated bins)
- Fusion: Late fusion via learned attention weights across three branches
- Classification Head: Fully connected layers with dropout (0.4) + Softmax

Training Strategy:

Phase1: Supervised Pre-training: Fine-tune EfficientNet-B4 backbone on combined spice dataset (lr=1e-4, AdamW, cosine annealing, 30 epochs).

Phase2 : Contrastive Fine-tuning: Apply SupCon loss on hard-negative spice pairs (e.g., cumin/caraway, turmeric/mustard powder) for 10 epochs.

Phase3: Full Model Training: End-to-end training with fusion module (10 epochs, lr=1e-5)

Tools & Frameworks: Python 3.10, PyTorch 2.x, Torchvision, scikit-learn, OpenCV, Albumentations (augmentation), Grad-CAM (interpretability), Weights & Biases (experiment tracking), HuggingFace Hub (model sharing).

Evaluation Metrics:

- Top-1 and Top-5 Classification Accuracy

- Macro-averaged F1 Score (handles class imbalance)
- Confusion Matrix — focus on hard-negative spice pairs
- Per-class Precision & Recall
- Inference time (ms/image) for deployment feasibility

Baseline Comparisons:

1. ResNet-50 (fine-tuned, image features only)
2. EfficientNet-B4 (fine-tuned, image features only)
3. ViT-Base (Vision Transformer, fine-tuned)
4. Traditional ML: SVM on HOG + Color Histogram features

All baselines trained under identical data splits and augmentation for fair comparison.

Ablation Studies:

A1-Component Ablation: Image-only vs. +Texture branch vs. +Color branch vs. Full fusion model

A2-Contrastive Loss Ablation: With vs. without Phase 2 contrastive fine-tuning

A3-Augmentation Ablation: No aug vs. standard aug vs. spice-specific aug

A4-Backbone Ablation: EfficientNet-B0/B2/B4 vs. ResNet-50 vs. MobileNetV3

A5-Background Removal: With SAM segmentation vs. raw images

Interpretability Analysis: Grad-CAM visualizations on misclassified pairs to understand model failure modes. Saliency maps to confirm the model focuses on discriminative spice texture/color rather than background artifacts.